Nobody doesn't like intergalactic space, but it can be hazardous to your health if you choose to go there. Let's ignore the fact that you would freeze to death as your warm body tried to reach equilibrium with the 3-degree temperature of the universe. And let's ignore the fact that your blood cells would burst while you suffocated from the lack of atmospheric pressure. These are ordinary dangers. From the department of exotic happenings, intergalactic space is regularly pierced by super-duper high-energy, fast-moving, charged, subatomic particles. We call them cosmic rays. The highest-energy particles among them have a hundred million times the energy that can be generated in the world's largest particle accelerators. Their origin continues to be a mystery, but most of these charged particles are protons, the nuclei of hydrogen atoms, and are moving at 99.9999999999999999 percent of the speed of light. Remarkably, these single subatomic particles carry enough energy to knock a golf ball from anywhere on a putting green into the cup.

Perhaps the most exotic happenings between (and among) the galaxies in the vacuum of space and time is the seething ocean of virtual particles—undetectable matter and antimatter pairs, popping in and out of existence. This peculiar prediction of quantum physics has been dubbed the "vacuum energy," which manifests as an outward pressure, acting counter to gravity, that thrives in the total absence of matter. The accelerating universe, dark energy incarnate, may be driven by the action of this vacuum energy.

Yes, intergalactic space is, and will forever be, where the action is.